

Talker change detection: A comparison of human and machine performance

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1 The automatic analysis of conversational audio remains difficult, in part due to the
2 presence of multiple talkers speaking in turns, often with significant intonation vari-
3 ations and overlapping speech. The majority of prior work on psychoacoustic speech
4 analysis and system design has focused on single-talker speech, or multi-talker speech
5 with overlapping talkers (for example, the cocktail party effect). There has been much
6 less focus on how listeners detect a change in talker or in probing the acoustic fea-
7 tures significant in characterizing a talker’s voice in conversational speech. This study
8 examines human talker change detection (TCD) in multi-party speech utterances us-
9 ing a novel behavioral paradigm in which listeners indicate the moment of perceived
10 talker change. Human reaction times in this task can be well-estimated by a model
11 of the acoustic feature distance among speech segments before and after a change in
12 talker, with estimation improving for models incorporating longer durations of speech
13 prior to a talker change. Further, human performance is superior to several on-line
14 and off-line state-of-the-art machine TCD systems.

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I. INTRODUCTION

Everyday speech communication involves more than extracting a linguistic message¹. Listeners also track paralinguistic indexical information in speech signals, such as talker identity, dialect, and emotional state². Indeed, in natural speech communication, linguistic and indexical information are likely to interact since conversations typically involve multiple talkers who take turns of arbitrary duration, with gaps on the order of only 200 ms³. On the listener’s side, the perception of conversational speech demands quick perception of talker changes to support communication.

Perceptual learning of talker identity enhances speech intelligibility in both quiet⁴ and acoustically-cluttered environments^{5,6}. This suggests that sensitivity to talker attributes affects speech recognition both in clear speech and under adverse listening conditions. Further, talker dependent adaptability in perception can be induced from exposure to just a few sentences⁷. These benefits hint at listeners’ ability to track talkers in conversational speech, even in the absence of visual or spatial cues.

Detecting a change in talker would seem to rely upon an ability to track regularities in the perceived features specific to a voice, and to detect changes from these upon a talker change. Lavner et al. (2009)⁸ suggest that the talkers are identified by a distinct group of acoustic features. Yet, Sell et al. (2015)⁹ argue that a combination of vocal source, vocal tract, and spectro-temporal receptive field¹⁰ features fail to explain perceived talker discrimination in a listening test with simple single-word utterances. In a similar way, Kimberly et al. (2003)¹¹ have described inattention to talker changes in the context of listening for comprehension as

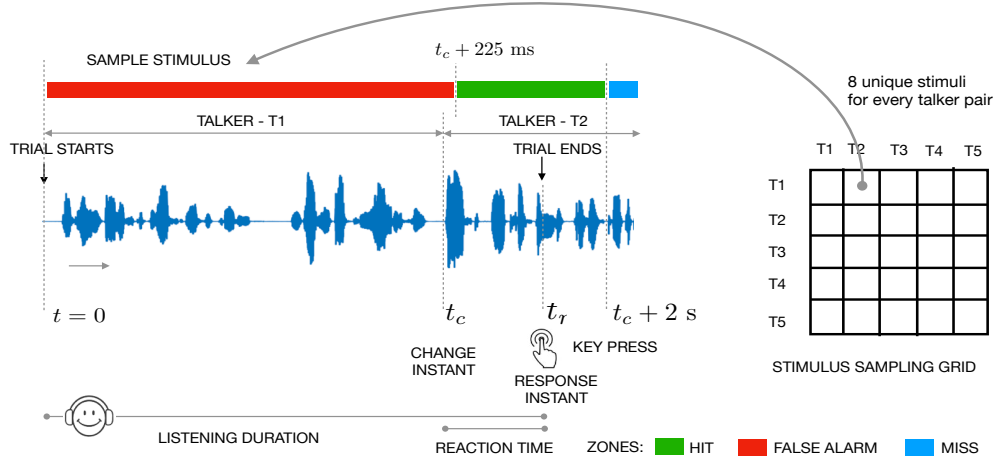


FIG. 1. (color online) Illustration of the proposed talker change detection (TCD) paradigm used in the present listening test study.

a form of talker change deafness¹². They suggest that voice information is not continuously monitored at a fine-grain level of acoustic representation, and conversational expectations may shape the way listeners direct attention to voice characteristics and perceive differences in voice. In fact, Neuhoﬀ et al. (2014)¹³ found improved voice change detection when the language is unfamiliar to the listener, suggesting that there may be interactions between linguistic and indexical information.

Acknowledging that conversational demands¹⁴ in natural speech will often shift attention toward acoustic features that signal linguistic rather than indexical information, listeners' ability to detect talker changes does suggest that they track the variability in acoustics features associated with a talker's voice. Yet, despite the importance of indexical characteristics of speech to communication, quite little is known about the nature of the detailed acoustic features across which talkers diﬀer, the distributions of information characterizing diﬀerent talkers along these acoustic features^{15,16}, and the listeners' ability to detect a change in

talker. This is especially true for fluent, connected speech, as opposed to isolated words. In this paper, we aim to advance understanding of the information human listeners use to track the change in talker in continuous multi-party speech. We first develop and test a novel experimental paradigm to examine human talker change detection (TCD) performance. We next model the listeners’ reaction time (RT) to respond to a talker change in relationship to multiple acoustic features as a means of characterizing the acoustic feature space that listeners may track in a voice. We then relate these human perceptual results with performance of state-of-the-art on-line and off-line machine systems implementing TCD.

A. Reaction time as a measure of TCD

We developed a novel paradigm with the goal of obtaining a continuous behavioral measure of listeners’ ability to detect a change in talker across relatively fluent, continuous speech. Each stimulus was composed of two concatenated 9 – 14 s utterances sourced from audio books, and spoken by either a single male talker or two different male talkers. The utterances were always drawn from different stories, or parts of a story, so that semantic continuity did not provide a clue to talker continuity. Listeners responded with a button press upon detecting a talker change, thus providing a continuous reaction time measure of how much of an acoustic sample was needed to detect a change in talker. Figure 1 provides an illustration of the paradigm. To the best of our knowledge, this is the first application of a RT change-detection approach to examine human TCD performance.

Past studies have used RT to analyze perception of simpler acoustic attributes. For example, studies of tone onset detection¹⁷ and broadband sound onset^{18,19} have reported

an inverse relationship between RT and stimulus loudness / spectral bandwidth. Studies guiding the design of warning sounds^{20,21} have shown faster detection of natural, compared to synthetic, stimuli. Particularly relevant to the present study, recent research characterizes human listeners’ ability to track distributional noise statistics in an audio stream by asking listeners to report a change in statistics with a button press²². Detection of a change was facilitated when listeners heard longer noise samples. This prior work illustrates the promise of the RT to detect a change as a means by which to examine how listeners build a model of incoming sound regularities, although speech signals are inherently more non-stationary and the distributional acoustic information that contributes to talker identity is not well understood.

B. Comparing machine and human performance in TCD

With an increasing repository of conversational audio data^{23,24}, automatic detection of talker change is considered to be an essential preprocessing in machine speech recognition²⁵. For example, the availability of timestamps corresponding to talker change instants in a recording benefits both speaker identification²⁶ and speech recognition²⁷ tasks. Recent results from the machine systems literature on TCD^{28–32} suggest that talker change detection is difficult. The state-of-the-art in TCD can be categorized into two approaches: metric-based and classification-based. The metric-based approach relies on computing a distance, in some feature space, between successive short-time segments (such as 25 ms segments, with successive shifts 10 ms) of a speech recording. A talker change is flagged upon detection of a distance that exceeds a preset threshold. The accuracy of TCD in metric-based approaches

is dependent on the choice of features (such as vocal tract features^{33,34} and vocal source features^{30,35–37}), segment duration (usually $> 3 - 5$ s), and the distance metric (such as likelihood ratio³⁸, Bayesian information criterion (BIC)^{33,39}, or improved BIC⁴⁰). An alternative is the classification-based approach whereby a binary classifier is trained to classify short-time speech segments as talker change or no talker change. The accuracy of the approach is dependent on the feature space, the complexity of the classifier (such as support vector machines⁴¹, neural networks³⁰, and deep neural networks^{31,42}), and the amount of training data. In the present work, we evaluate the performance of a few state-of-the-art machine systems across the same stimuli and with the same performance metrics as used in the human perceptual paradigm. These comparisons may help in identifying whether there are performance gaps in human and machine TCD across fairly continuous, fluent speech.

II. METHODS

A. Participants

A total of 17 participants in the age group of 18 – 36 (university students and one staff member, median age 26) took part in the listening test. All listeners reported normal hearing with good fluency in speaking and understanding English. All participants provided informed consent to participate in the study, and the study protocol was approved by the Carnegie Mellon University Institutional Review Board and the Indian Institute of Science Review Board.

B. Stimuli

The stimuli were composed of concatenated utterances from two talkers, talker T_x and talker T_y . The utterances were taken from audio books drawn from the LibriSpeech corpus⁴³, a public-domain corpus of about 1000 hours of audio data by approximately 1000 distinct talkers. Each trial involved a stimulus that was a concatenation of two utterances drawn from the same, or two different, talkers. The utterances corresponded to sentences read out in the audio book, and featured natural speech intonation and a rise and fall in speech envelope at the start and end. The sentences were chosen randomly from the respective talker's audio book. To avoid an obvious talker change detection due to gender attributes, we chose all

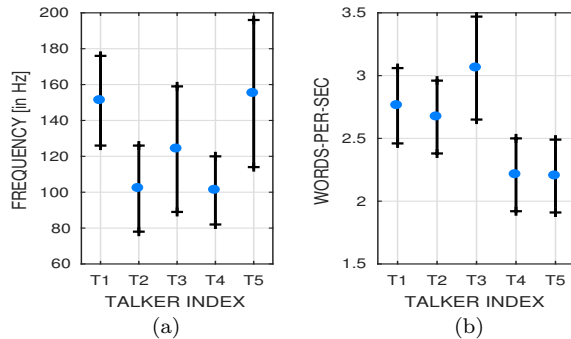


FIG. 2. A comparison of talker attributes with respect to (a) fundamental frequency variation, and (b) word speaking rate. The vertical bars indicate one standard deviation spread around the mean value.

male talkers. Based on an informal pilot experiment aimed at finding a set of perceptually separable voices, we chose five talkers from the corpus (IDs 374, 2843, 5456, 7447, 7505) for the listening test stimulus design (here referred to as T1, T2, etc.). The average fundamental frequency (estimated using STRAIGHT⁴⁴) and speaking rate expressed as words spoken per

second (estimated from the audio book transcripts) of the five talkers are depicted in Fig. 2, with significant overlap across talkers to make for challenging TCD.

To make a stimulus, talker T_x was chosen from the list of N talkers, and a sentence utterance was retrieved from the corresponding talker’s audio book. A short utterance from another talker T_y was chosen, and this was concatenated to the utterance from T_x . As the utterances were natural speech, there were natural pauses. Owing to this, the silent interval between T_x ’s end and T_y ’s start after concatenation was random and ranged from 200 – 1000 ms. In any stimulus, speech corresponding to T_x was between 5 – 10 s and that corresponding to T_y was 4 s. A sample stimulus is shown in Fig. 1.

For each pair of T_x - T_y talkers there were $M = 8$ unique stimuli. The stimulus set can be represented as sampled from a grid, as shown in Figure 1. This resulted in a total of $M \times N^2 = 200$ distinct speech stimuli, each 9 – 14 s in duration.

C. Protocol and apparatus

A graphical user interface (GUI) for stimulus presentation was made using Gorilla¹, a software platform for designing behavioral science tests. A test session comprised three tasks carried out in sequence, namely, Task-1, 2 and 3. Task-1 measured RT for noise-to-tone detection and Task-2 measured RT for tone frequency change detection (see Supplementary material²). The acoustic attributes associated with a change in the stimuli in these first two tasks were easily recognizable. As a result, Task-1 and Task-2 served as benchmarks against which to compare human RT on Task-3, associated with TCD.

In each task, listeners were instructed to press a button (**space bar**) immediately upon detection of a change in the stimulus. The audio stopped after the button press and visual feedback indicating $\checkmark$ (or X) for correct (or incorrect) detection appeared immediately. Participants were seated in sound-attenuated booths wearing Sennheiser headphones³(with

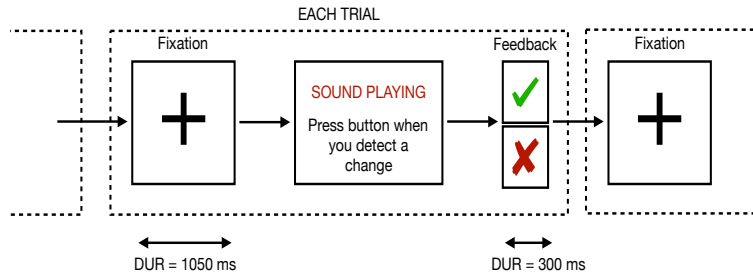


FIG. 3. An illustration of a listening test trial.

flat spectral response from 60 – 8000 Hz), with diotic stimulus presentation. In order to prevent fatigue, participants were allowed to take breaks after blocks of 25 trials. Each participant listened to a few (8 – 10) different Task-3 TCD stimuli (not used in the test) to become conversant with the paradigm. On average, the complete test session was 45 mins (with Task-1, Task-2, and Task-3 taking 5, 10 and 30 mins, respectively).

D. Performance measures

For each trial in which there was a button press, the RT for change detection was obtained as the difference between the response instant (denoted by t_r) and the ground-truth acoustic change instant (denoted by t_c), that is, $RT = t_r - t_c$. An illustration is provided in Figure 1. The lower limit for RT for change perception in sound attributes is on the order of $RT < 250\text{ ms}$ ¹⁹. Hence, RTs in the range 0 – 250 ms are likely to be associated with speech heard

prior to the change instant t_c . The upper bound on RT (2000 ms) was chosen based on prior research²².

We analyzed the hits, misses, and false alarms (FA) in the responses. The 200 trials in Task-3 per subject were categorized into two pools for analyses:

Pool A, involving trials with T_x a different talker from T_y (two-talker trials) and either $RT > 225$ ms or no button press.

Pool B, involving trials with $T_x = T_y$ and trials with $T_x \neq T_y$ but $RT < 225$ ms. These all are single-talker trials (i.e the trials in which the subject's response was based on attention to only one talker).

From these pools of data, we defined the following detection measures:

- **Hit rate:** A hit corresponds to a trial in *Pool A* with $225 \text{ ms} < RT < 2000 \text{ ms}$. Hit rate is the ratio of number of hits to the number of trials in *Pool A*.

- **Miss rate:** A miss corresponds to a trial in *Pool A* with $RT > 2000 \text{ ms}$. Miss rate is the ratio of number of misses to the number of trials in *Pool A*. Note that the miss rate is $100 - \text{hit rate}$.

- **False alarm rate:** A false alarm (FA) corresponds to a trial in *Pool B* featuring a button press. False alarm rate is the ratio of number of FAs to the sum of trials in *Pool B* and *Pool A* (this equals 200).

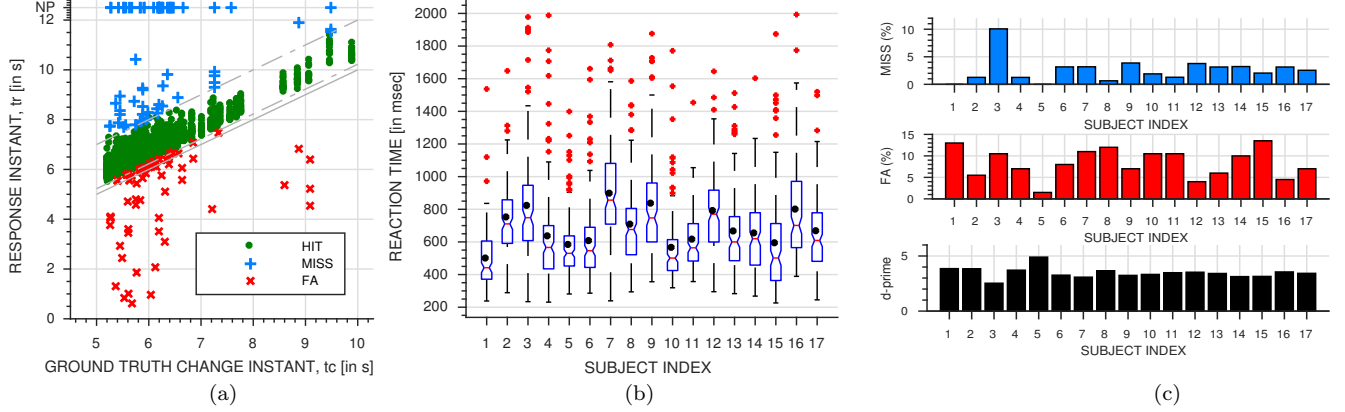


FIG. 4. (color online) (a) Illustration of human reaction time (RT) versus the ground-truth talker change instant (t_r vs t_c) across a total of 2720 trials (with $T_x \neq T_y$) over 17 subjects. The three inclined gray lines from bottom to top correspond to $t_r = t_c$, $t_c + 225$, $t_c + 2000$, respectively. NP stands for no button press. (b) Subject-wise summary using a boxplot of RTs in trials with hits. The black dots correspond to means. (c) Subject-wise miss and false alarm rates, and d-prime obtained from 200 trials for each subject.

III. RESULTS: HUMAN TCD EXPERIMENT

A. RT distribution

Figure 4(a) depicts the distribution of TCD reaction time t_r as a function of ground-truth talker change instant t_c for all trials which have a talker change. (taken from *Pool A* and *Pool B*). As seen, the majority (approx. 95%) of responses fall in the hit zone, that is, $t_c + 225 < t_r < t_c + 2000$ ms. Analyzing the hit trials from *Pool A*, the subject-wise RT summary is shown in Figure 4(b). Across subjects, the response time to detect a talker change tended to require mostly under a second of speech from the true change instant, with subject-dependent distributions of average RT and variability across quantiles. Analyzing

the detection parameters, the subject-wise hit, miss and FA rates are shown in Figure 4(c). The hit, miss, and false alarm rates averaged across all subjects were 97.38%, 2.62%, and 8.32%, respectively. Listeners performed the TCD task very accurately; the average d-prime across subjects was 3.48 (d-prime is defined as $\mathcal{Z}(\text{hit rate}) - \mathcal{Z}(\text{FA rate})$, where function $\mathcal{Z}(p)$, $p \in [0, 1]$, is the inverse of the cumulative distribution function of the Gaussian distribution)

The distribution of RT corresponding to hits from all subjects is shown in Figure 5. The non-Gaussian nature of the data is evident from the histogram and the normal probability plot. To improve the Gaussianity, we applied a log transformation ($\log_{10} RT$) on the RT data; the resulting distribution is shown in the same figure. We used this transformed data in the regression analysis, which is presented next.

B. Dependence of RT on speech duration

We examined the extent to which the duration of speech experienced prior to a talker change impacted TCD. We probed this using linear regression analysis on log-RT (averaged across subjects) versus speech duration before change instant t_c . As the stimulus set is composed of naturally spoken sentences from story books, the grid along speech duration is non-uniformly sampled in the dataset. Hence, we performed the regression analysis only for stimuli with $t_c < 7000$ ms, as beyond this t_c value, the grid was sparsely sampled. The result is shown in Figure 6. The high variability in log-RT may be attributed to the complicated feature space associated with speech signals. This is unlike what is observed in tone frequency (or noise-to-tone) change detection for which the variability in RT correlates with the simple

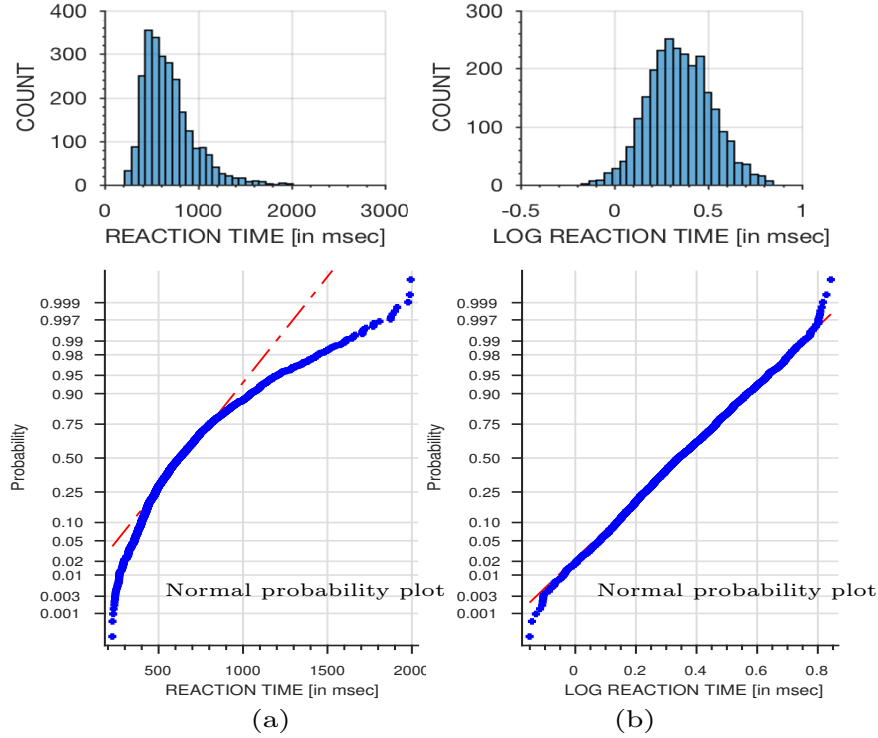


FIG. 5. (color online) Illustration of the distribution (obtained as a histogram) of the RT data for trials on which there was a hit for (a) raw RT data and (b) log-transformed RT data to improve the fit to a normal distribution.

210 spectral difference (see Supplementary material and also²²). Despite considerable variability,
 211 we observe a trend such that TCD is slower with shorter samples of speech from the initial
 212 talker (a decrease in the t_c value). Next, we attempt to model RT as a function of different
 213 acoustic features extracted from the stimulus before and after change instant to understand
 214 the acoustic dimensions listeners may track in TCD.

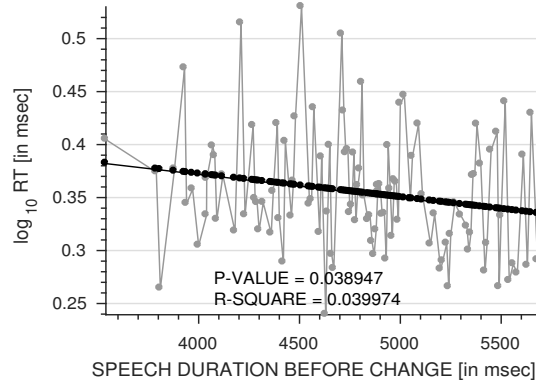


FIG. 6. Dependence of average RT on speech duration before the change instant. The black line is the linear regression fit.

C. Modeling RT with acoustic features

Past studies^{48,49} have used RT modeling to reveal visual input features associated with object recognition. Motivated by this, here we model RT to detect a change in talker across hit trials as a function of the difference in the acoustic feature space between speech segments sampled before and after the change instant. We consider a collection of acoustic features that may be important in tracking voice identity, and examine their ability to estimate human participants' TCD RT.

The approach is illustrated in Figure 7(a,b). Let D_b and D_a denote segments of the stimulus before and after change instant t_c , respectively. We hypothesize that a listener estimates acoustic features from these segments, summarizes them and compares the summary using a distance measure. The resulting distance serves as strength of evidence for change detection, and by Piéron's law⁵⁰ this should impact the RT. The greater the distance, the faster listeners may detect a talker change and thus the smaller the RT value. We assume

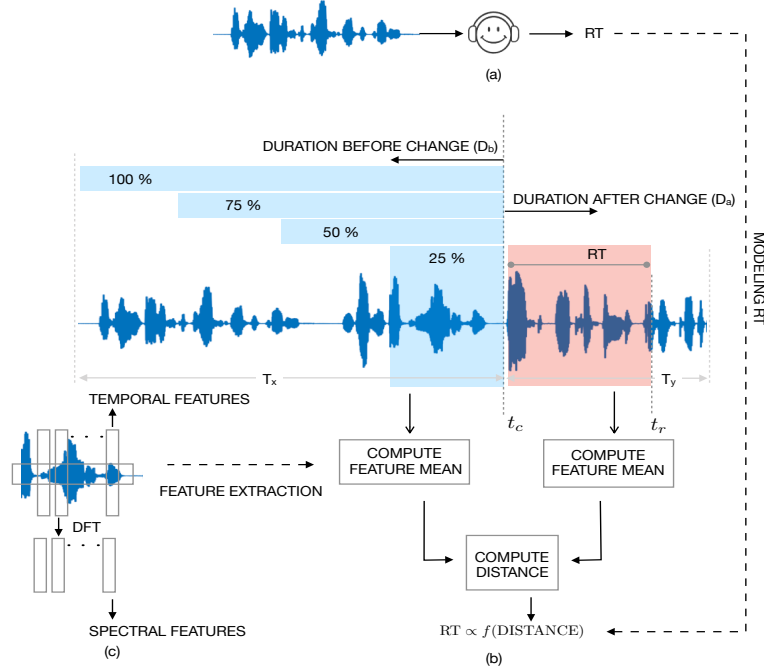


FIG. 7. (color online) Proposed approach to model RT using acoustic features before and after change instant.

D_a corresponds to the segment from t_c to t_r . For D_b , we model different segment durations before the change instant (that is, 25%, 50%, 75%, and 100% of t_c duration segment, before change instant t_c).

1. Feature computation

A wide range of acoustic features can be computed from D_b and D_a segments. Here, we consider the 9 sets of features described in Table I. These are chosen to sample from a range of possible temporal and spectral acoustic features (illustrated in Figure 7(c)). From the perspective of speaker attributes, the feature set can be further grouped into those capturing pitch (F0), vocal tract formants (LSFs, MEL, MFCCs), rate of temporal variation of vocal tract features (temporal derivatives⁵¹ of MFCCs, namely, MFCC-D and MFCC-

TABLE I. Acoustic features used in the regression model analysis.

FEATURE SET	FEATURES	TYPE	DIMENSION	TIME SCALE
F0	Fundamental Frequency	Spectral	1×1	25 ms
LSF	Line spectral frequencies	Spectral	10×1	25 ms
MEL	Mel-spectrogram	Spectral	40×1	25 ms
MFCC	Mel-frequency cepstral coefficients	Spectral	12×1	25 ms
MFCC-D	First-order temporal derivative of MFCCs	Spectral	12×1	25 ms
MFCC-DD	Second-order temporal derivative of MFCCs	Spectral	12×1	25 ms
ENGY-D	Derivative of short-time energy	Temporal	1×1	25 ms
PLOUD	Loudness strength, sharpness, and spread	Spectral	3×1	25 ms
SPECT	Spectral flatness, Spectral flux, Spectral roll-off,	Spectral	8×1	25 ms
	Spectral shape, Spectral slope			

DD), perceived loudness (PLOUD⁵²), spectral timbre (SPECT), and the rate of temporal variation in short-time energy (ENGY-D). These features are computed every 10 ms with *Hanning* windowed short-time segments of 25 ms. All features were extracted using the *Yaafe*⁵³ Python package, an efficient open-source code library for speech and audio analysis.

2. Regression on feature distance

For each feature set, we summarized the segments D_b and D_a using the mean of the features in each segment. The PLOUD, and SPECT feature set were characterized by a combination of different features. Hence, we mean- and variance-normalized these feature sets over the whole duration prior to segment-wise mean computation. Following this, we computed the Euclidean distance between the obtained means. Owing to significant variability in RT across subjects (see Figure 4(b)), we modeled each subject's RT separately.

We defined the number of trials with a hit for the p^{th} subject to be denoted by N_p . Corresponding to these trials, we have N_p RTs, and for each RT we compute a distance

252 between the mean features extracted from segments D_b and D_a . There are 9 such distances
 253 based on the choice of features, and we denote these by d_k , $k = 1, \dots, 9$, that is, one distance
 254 for each feature set. To evaluate the impact of the feature distances on RT to detect a talker
 255 change, we perform a linear regression on feature distances to estimate the RT using a
 256 regression model for the k^{th} feature set

$$\underbrace{\begin{bmatrix} \log RT_1 \\ \log RT_2 \\ \vdots \\ \log RT_{N_p} \end{bmatrix}}_{\mathbf{r}} = \underbrace{\begin{bmatrix} 1 & d_{k,1} \\ 1 & d_{k,2} \\ \vdots & \vdots \\ 1 & d_{k,N_p} \end{bmatrix}}_{\mathbf{D}} \underbrace{\begin{bmatrix} w_0 \\ w_k \end{bmatrix}}_{\mathbf{w}} \quad (1)$$

257 where w_0 and w_k are the model parameters representing the mean RT and slope of the
 258 regression line, respectively, and RT_i and $d_{k,i}$ denote the RT and feature distance in the i^{th}
 259 trial, respectively. We solve for the model in (5) using minimum mean square error (MMSE).
 260 That is,

$$\hat{\mathbf{w}} = \min_{\mathbf{w}} \|\mathbf{r} - \mathbf{D}\mathbf{w}\|_2 = (\mathbf{D}^T \mathbf{D})^{-1} \mathbf{D}^T \mathbf{r} \quad (2)$$

$$\hat{\mathbf{r}} = \mathbf{D} \hat{\mathbf{w}}. \quad (3)$$

261 The MMSE optimized values of w_k for all subjects and for the 9 features sets are shown in
 262 Figure 8(a). These are obtained for D_b set to 100% of t_c . For a majority of subjects, w_k is
 263 negative for all the feature sets. This signifies that, on average, RT decreases with increased
 264 feature distance. Some feature sets have a greater negative slope than others. For example,
 265 the slope is maximally negative for MFCC-D and MFCC-DD feature sets. To quantify the

modeling performance, we used the r-square measure⁵⁴, computed as

$$\text{r-square} = 1 - \frac{\|\mathbf{r} - \hat{\mathbf{r}}\|_2^2}{\|\mathbf{r} - \bar{\mathbf{r}}\|_2^2} \quad (4)$$

where $\bar{\mathbf{r}}$ is the mean of elements in $\mathbf{r}$. The r-square is also referred to as the “explained variance” by the model; a value close to 100% indicates good modeling performance; that is, the proposed model is able to better explain the observed variance in the data.

Figure 8(b) shows the obtained r-square (shown in % as explained variance) for different feature sets. For clarity of presentation, the depicted percentage is the average percentage across all subjects. At the level of individual features, the MFCC-D outperforms all other feature sets. Moreover, a majority of feature sets fall below 10%, thereby failing to explain a significant portion of the variance in RT.

To examine combinations of features sets, we performed a multiple linear regression by combining the feature distances from all feature sets as follows,

$$\underbrace{\begin{bmatrix} \log RT_1 \\ \log RT_2 \\ \vdots \\ \log RT_{N_p} \end{bmatrix}}_{\mathbf{r}} = \underbrace{\begin{bmatrix} 1 & d_{1,1} & \dots & d_{9,1} \\ 1 & d_{1,2} & \dots & d_{9,2} \\ \vdots & & & \\ 1 & d_{1,N_p} & \dots & d_{9,N_p} \end{bmatrix}}_{\mathbf{D}} \underbrace{\begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_9 \end{bmatrix}}_{\mathbf{w}} \quad (5)$$

This gave the best r-square (see Figure 8(b)). A p-value (from hypothesis testing) illustration depicting the relevance of each feature set in the multiple linear regression is shown in Figure 8(c). The MFCC-D and MFCC-DD are most relevant, $p < 0.05$, across all subjects. A scatter plot of RT versus estimated RT , pooling all subjects’ trials, is shown in Figure 8(d).

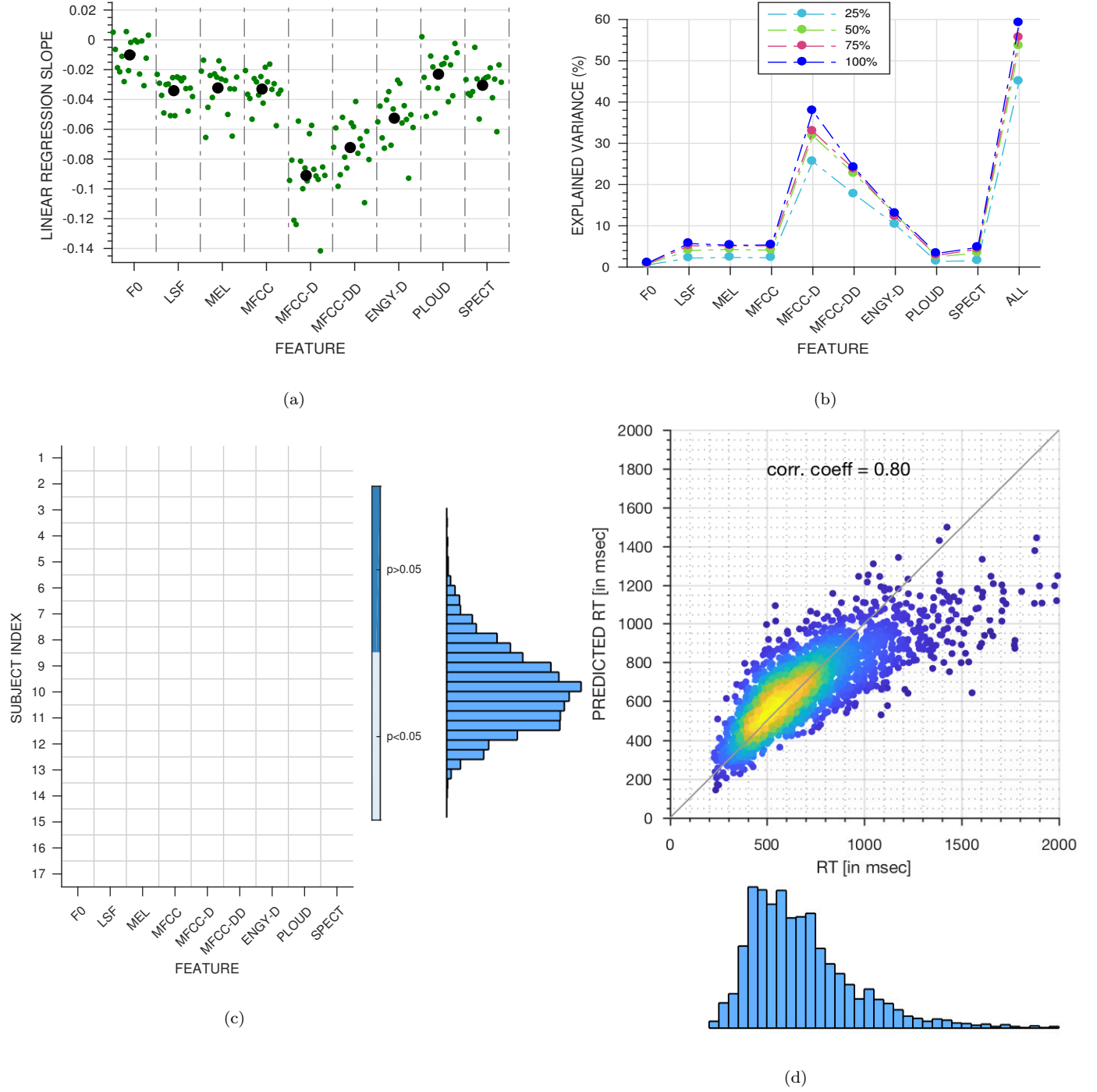


FIG. 8. (color online) (a) Linear regression slope indicating the performance of individual feature sets in accounting for the variance in human TCD reaction times. (b) The percentage of explained variance (r-square) for the linear regression model, averaged across subjects. (c) Illustration of significance (p-value) of different feature sets in multiple linear regression. (d) Scatter plot (polling all subjects' trials) of true RT versus estimated RT obtained from a multiple linear regression model computed across all feature sets.

The Pearson correlation coefficient amounted to 0.8, indicating a good overall estimation accuracy. Also, it can be seen that a major portion of true RT fall in the range 400–1000 ms, with few trials with $RT > 1000$ ms. In this range, the estimation is also concentrated along the $y = x$ line.

We also tested the modeling performance with decreasing duration of segment D_b , that is duration set to 75, 50, or 25% of t_c duration before the change instant (a shown in Figure 7(b)). The result shown in Figure 8(b) depicts that using 100% of the segment duration better models human performance, with systematic decreases as the sample of the talker’s speech decreases in duration.

D. Talker pair-wise analysis

To analyze the variability in TCD performance across the talker pairs, we examined talker-wise performance in TCD RT (see Figure 9). Most of the talker pairs have the average RT (computed across subjects) in the same range, except $T_1 - T_5$ and $T_5 - T_1$. Also, the miss rate was found to be higher for these pairs of talkers. This suggests that these pairs may be overlapping a lot in the perceived talker space. Comparing the FA rate, averaged across subjects, talker T_3 had the highest FA rate.

IV. MACHINE SYSTEM FOR TCD

We evaluated the performance of three machine TCD systems on the same stimulus materials used in the human TCD experiment. The first system was an adaptation of a state-of-the-art diarization system, designed to segment audio into distinct talker segments based on

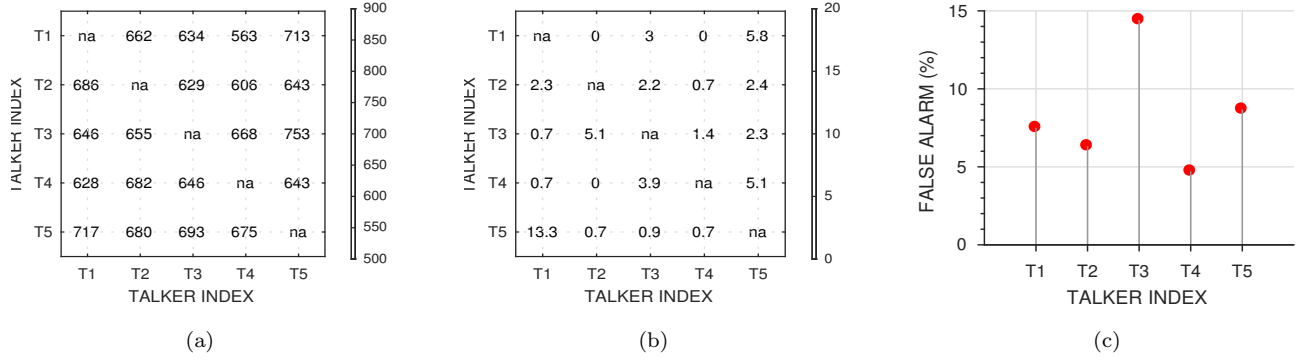


FIG. 9. (color online) Dependence of: (a) average RT on talker pairs (T_x-T_y), (b) average miss rate on talker pairs (T_x-T_y), and (c) average false alarm rate across talkers.

i-vector and probabilistic linear discriminant analysis (PLDA)⁵⁵. Subsequently, the talker change instants can be obtained as segment boundaries. Traditionally, these systems operate across a whole audio file. We refer to this mode of operation as off-line TCD.

In contrast to off-line machine systems, listeners in the human TCD task did not listen to the whole audio file. Instead, they performed an on-line change detection, pressing the button as soon as a talker change was perceived. To better model this aspect of human listening, we implemented an on-line variant of the off-line machine system. Here, the system sequentially operated on segments of increasing duration starting with an initial 1 s segment for a given audio file. The sequential operation was stopped as soon as the second talker was detected, with this instant corresponding to the response instant for a talker change in machine recognition. An illustration of the approach is shown in Figure 10. As we hypothesized for human listeners, the on-line system uses the acoustic features extracted from onset until the current segment duration in making a decision for talker segmentation.

A. On-line and Off-line diarization-based TCD systems

We first segmented an audio file into small (750 ms) and temporally-overlapping segments (temporal shift of 500 ms). Each short segment was transformed into an i-vector representation⁵⁷. All pair-wise distances between the short-segment i-vectors were computed using a PLDA based scoring approach. In the literature, PLDA based scoring has been shown to impart robustness against small duration recordings (5 – 10 s) and variations in talking style⁵⁸, and this suit the present stimulus set. Using the PLDA scores, agglomerative hierarchical clustering was performed to identify short-segments belonging to the same talkers and subsequently to merge the short-segments of the same talker. The obtained output was a segmentation of the input audio file into distinct single talker segments.

We developed the diarization system in the following two modes:

Off-line Diarization Here, the diarization was performed on the complete audio file in one pass.

On-line Diarization Here, instead of doing diarization across the complete audio file in one pass, we began with an input of 1 s and then sequentially increased it by 1 s, until two talker segments were detected or the end of file was reached (illustrated in Figure 10).

The complete system setup was developed using the Kaldi toolkit⁵⁹. This involved training the i-vector extractor based on a universal background model composed of a 512-component Gaussian mixture model with a diagonal covariance matrix and trained on the LibriSpeech corpus⁴³. The system used 12-dimensional MFCC features, obtained from successive 25 ms (with temporal shifts of 10 ms) short-time segments derived from the audio files. The

MFCC features were mean- and variance-normalized using a 3 s running window. The i-vector representations were 128-dimensional. The audio files corresponding to talkers used in the listening test were removed from the training dataset.

Since the extraction of i-vectors involves an overlapping shift, the system can flag a change instant before the actual ground-truth change instant. In order to account for this look-ahead, for these systems a tolerance window $\delta = 500$ ms was given and the RT corresponded to a hit if $t_c - \delta < RT < t_c + 2000$ ms. The operating point for the system can be tuned by varying the segment clustering threshold. Thus, each operating point had its own miss and false-alarm-rates. Hence, we generated a detection error trade-off curve for each system. These are shown in Figure 12.

B. IBM Watson speech-to-text (STT) system

Along with an automatic speech recognizer (ASR), the commercially-available IBM Watson STT system also includes a follow-up TCD system. Built as a one-of-its-kind approach⁶⁰, the TCD system makes use of word boundaries from the ASR output in addition to acoustic feature information. For each acoustic segment corresponding to a pair of contiguous words, two separate Gaussian models are fit using MFCC features from left and right of the detected word boundary, respectively. A talker change is flagged based on the BIC algorithm or T^2 -criterion. Use of word annotations from ASR reduces false alarms in change detection within a word and in non-speech regions. We used the off-the-shelf implementation of this system available as a Web API. This system is pre-tuned and provides only one operating point. Like the diarization system, this system also was found to often give a change instant

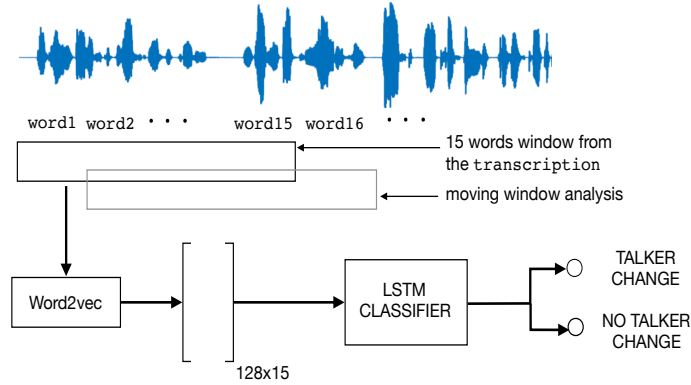


FIG. 11. Illustration of text-based context change classification.

before the ground-truth change instant. This may be due to the larger temporal context used in the ASR module. On post analysis for hit rate computation, the detected change instant was found to lie in $t_c - \delta < RT < t_c + 2000$ ms, with $\delta = 200$ ms. We considered all these responses as correct detections in hit rate computation. The resulting miss and false alarm rates are shown in Figure 12.

C. TCD based on textual features

Although we hypothesize that acoustic feature distributions are likely to play a major role in the listeners' model of talker identity, it is also likely that the listeners attended to the semantics of the words making the stimuli.

Designing an experimental paradigm that mitigates semantic contributions in order to evaluate the acoustic features that contribute to TCD is challenging. The present paradigm takes the approach of introducing a *semantic* change on every trial. Hence, listeners cannot rely on a semantic change as a reliable cue for TCD as some trials feature no talker change. The challenge to disentangle the contribution of text/content versus acoustic features led us

to examine a machine equivalent of TCD reliant upon only the information conveyed by the text of our stimulus material^{61,62}. The proposed text-based TCD system is shown in Fig. 11. Using the transcripts from the LibriSpeech corpus, we trained a Word2Vec model with the Gensim Python package⁶³. The model represents every word with 128-dimensional vector. This allows representing a sentence as a sequence of vectors obtained from the constituent words. We built a talker change detector by analyzing the semantic similarity among sets of words via analysis of the vector representations.

Specifically, the system is a classifier that takes N consecutive words as input and outputs a class label C_0 if all words are by a single talker, and a class label C_1 otherwise. The value of N in this experiment was chosen after analyzing the average number of words in a sentence spoken by the first talker as well as the average number of words spoken by the second talker within 2 s. Using features from 15 consecutive words from the corpus of talkers reading audio books, we generated training examples for the two classes. The training set comprised of two different talkers (label C_1) was created by taking the last 10 words of a sentence from the transcript of an audio book read by one talker and the first 5 words from the transcript of an audio book read by another talker. For the set of examples with no talker change (class C_0), the dataset was created by taking 10 words from the end of one sentence and the first 5 words from the beginning of the following sentence, both drawn from transcripts of the audio book read by one talker. This dataset was created excluding the talkers in the listening set. Using the training examples, we train a Long Short-Term Memory Network (LSTM) to predict talker change based on context. The model was designed using the Keras toolkit⁶⁴ comprising of one LSTM layer with 512 cells followed by three dense layers with

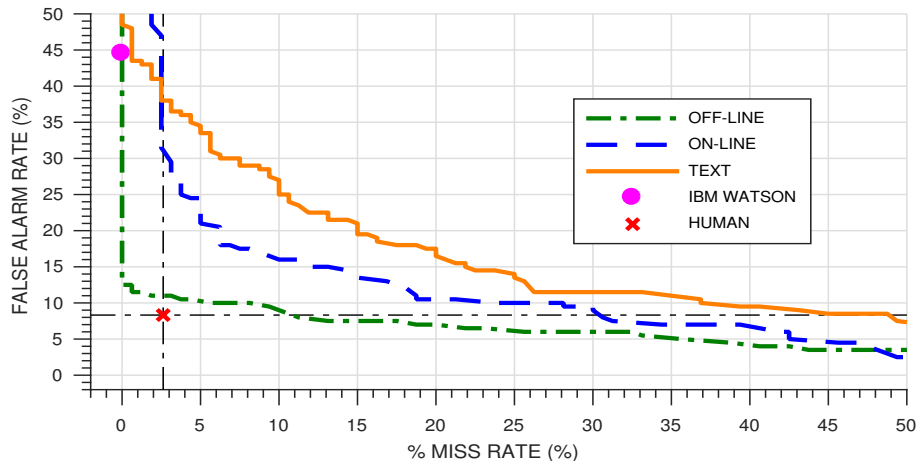


FIG. 12. (color online) Detection Error Trade-off (DET) curve for all the TCD systems evaluated on the stimuli set used in the listening experiments. The IBM Watson system is a single operating point. The human score is also marked in this figure for reference.

1024, 512 and 256 neurons, respectively. The test corpus was made from the transcripts of
the sentences used in the listening test. Each sentence was input as a series of $N = 15$ words,
to reflect the training set input, shifting a word by one to capture all words. A stimulus is
marked as a change when it consists of at least one word from C_1 . For computing detection-
error-trade-off curves, a threshold was used on the posteriors from the LSTM model. The
resulting performance is shown in Figure 12.

V. DISCUSSION

Summarizing the findings, the results from the experiments make three primary contribu-
tions towards understanding human and machine TCD performance.

A. Human performance for TCD

Human listeners performed the TCD task with very high accuracy, averaging only a 2.62% miss rate and a 8.32% false alarm rate. The presence of false alarms signifies that human listeners sometimes reported a talker change when there was none. This can be partly attributed to varying stress, intonation, and speaking rate exhibited by each talker during the course of reading the story book. At the same time, the high accuracy contrasts with some prior studies reporting high rates of “change deafness” to a change in talker¹¹. However, it is important to note that the present task required listeners to direct the attention to detect a change in talker rather than to the comprehension alone. As suggested in prior research¹¹, conversation expectations and task are likely to influence human listeners’ direction of attention to fine-grain details of voice. At least in the context of an overt task requiring TCD, the present results demonstrate human listeners’ ability to track acoustic features across talkers and to differentiate the dimensions most relevant to detecting a change among male voices. Here, in the interest of investigating the issue in a controlled task, we examined read speech from audio books. The same approach might be applied to other stimulus sets capturing even more natural conversational speech, although in these cases covariation with context, semantic continuity, and other factors would be likely to complicate attempts to understand listeners’ ability to track distributional information across acoustic dimensions that are related to talker identities.

Speculating from average RT to detect a change, the perceptual load for TCD appears to be greater than for simpler acoustic change detection scenarios such as noise-to-tone change

or tone frequency change in the same listeners (experiments reported in supplementary material). Pooling trials across listeners, the average RT for change detection was 680 ms (std. dev.= 274 ms), which was close to twice the average RT for a noise-to-tone change detection, and falls within the average RT associated with a tone frequency change detection task (done with different magnitudes of change). We speculate that these differences may be attributed to recruitment of different auditory sub-processes for change detection in the context of speech and non-speech stimuli, specifically the need to accumulate distributional acoustic information across the more complex, multi-dimensional acoustic features that convey talker identity.

B. Estimating RT using a simple regression model on feature distances

We used a linear model to relate human listeners’ log RT to detect a talker change and acoustic feature distances across a set of acoustic features. A simple Euclidean distance measure between the mean of the acoustic feature measurement corresponding to speech segment before and after change instant was used. Interestingly, we found the Piéron’s law⁵⁰, stating decrease in RT with increase in “strength of evidence” (such as loudness or frequency difference for tone stimuli), to hold for distance computed in the feature space (negative slopes in Figure 8(a)) for talker change detection, as well. Quantifying the model performance in terms of the percent of the human performance variance explained, the best fit was obtained in the MFCC-D feature space, followed by MFCC-DD and ENG-Y-D features. The 12-dimensional MFCC representation derived from MEL representation is a spectrally smoothed representation of the short-time spectrum, preserving the spectral peaks corresponding to formants,

with minimal pitch information. The MFCC-D representation, derived from the temporal derivative of MFCC, captures the rate of variation in the spectrally smoothed representation. The improved model performance with the MFCC-D feature suggests that listeners are likely using the spectro-temporal variability in formant frequency space while attending to detect talker changes. We found a poor model fit with fundamental frequency (F0), often implicated in talker differences. This may be because there was considerable overlap in the fundamental frequency spread across our all-male set of talkers (shown in Figure 2). Likewise, it is notable that application of a multiple regression model using the 9 feature sets as independent variables improved the model performance significantly ($r\text{-square} \approx 0.6$). A significant contribution in this model came from MFCC-D and MFCC-DD ($p < 0.05$, see Figure 8(c)). This suggests that TCD involved tracking acoustic information across a quite complicated feature space. Visualizing the quality of estimated and true RT s (pooling from all subjects) we found a correlation of 0.8 (Figure 8(d)). A majority of true RT s fall in the range 400 – 900 msec, and a significant portion of this was mapped to the same range by the proposed model. Focusing on the duration of speech segment before change instant used in the model estimation showed a systematic improvement in performance with duration for all the features (Figure 8(c)). This suggests that TCD response is less predictable using local information around the change instant and likely the listener continuously builds a talker model while attending to speech.

C. Machine performance on the TCD Task

The off-line diarization system provided better performance than the on-line system machine TCD system. This can be attributed to improved clustering of the i-vectors into two clusters because of availability of more data after change instant compared to the on-line case. The text-based TCD system (IBM Watson) exhibited relatively poor performance, indicating that, for the stimulus set, the text features did not suffice for TCD. This provides assurance that our novel paradigm for assessing human TCD (in which a change in sentence context was introduced on each trial independent of a talker change) was sufficient to reduce any bias arising from the semantic content of the utterance before and after the change instant. The IBM Watson has a false alarm rate too high for successful use of the system in natural speech conversations, and underscores the importance of modeling the acoustic distributional characteristics of talkers voices in supporting successful diarization. In fact, documentation of the system does caution that the presence of speaking style variations in the utterances may lead to high false alarm rates. Comparing human and machine systems we found a considerable gap in performance, with humans significantly outperforming state-of-the-art machine systems (see Figure 12). In all, the present results highlight the complexity of the acoustic feature space that listeners must navigate in detecting talker change and underscores that machine systems have yet to incorporate the optimal feature set to model human behavior.

494 VI. CONCLUSIONS

495 The key contributions from the paper can be summarized as follows,

496 (i) Developing a novel paradigm for probing the human talker change detection (TCD)
 497 across short-duration natural speech utterances.

498 (ii) Characterizing human TCD performance using various parameters - reaction time
 499 (RT), hit-rate, miss-rate and false alarm-rate.

500 (iii) Building a simple linear regression based model that estimates the human RT in TCD
 501 using the distance between mean acoustic features from speech segments. This model
 502 revealed the significance of MFCC-D and MFCC-DD acoustic features in TCD.

503 (iv) Comparing and benchmarking the machine TCD system performance implemented
 504 using principles of speaker diarization and textual features with human TCD perfor-
 505 mance.

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716 ²See Supplementary material for supplementary experiments and results on change detec-
717 tion.

718 ³Sennheiser HD 215 II Closed Over-Ear Back Headphone with High Passive Noise Attenu-
719 ation.

720 ⁴<https://github.com/IBM-Bluemix-Docs/speech-to-text> (Last viewed: August 04,
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